# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME:** **Cryptography and Network Security**

**COURSE CODE:** **CSA 5113**

**COURSE OUTCOME COVERED**

**CO1**: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

**QUESTIONS: 5** **Total Marks: 100**

Annexure A

## CONCEPT MAPPING

***Code of Conduct:***

*I Gurusurya G 192311332_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Gurusurya G*
***Signature of the student:***

| *Q. No* | *Understanding of Concepts (25)* | *Experimental Design (30)* | *Use of Tools (20)* | *Analysis of Results (15)* | *Documentation (10)* | *Total Score (out of 100)* |
|---|---|---|---|---|---|---|
| *1* | / 5 | / 6 | / 4 | / 3 | / 2 | |
| *2* | / 5 | / 6 | / 4 | / 3 | / 2 | |
| *3* | / 5 | /6 | / 4 | / 3 | / 2 | |
| *4* | / 5 | / 6 | / 4 | / 3 | / 2 | |
| *5* | / 5 | / 6 | / 4 | / 3 | / 2 | |
| ***OVERALL RUBRICS VALUE*** | | | | | | |
| | / 5 | / 5 | / 4 | / 4 | / 2 | |
| | ***/ 25*** | ***/ 30*** | ***/ 20*** | ***/ 15*** | ***/ 10*** | ***/ 100*** |

Annexure A

# CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

## RUBRICS FOR CALCULATION:

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25 | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design | 30 | Well-planned, systematic implementation with optimal solution | Correct implementation with minor issues | Basic implementation with errors or inefficiencies | Incorrect or incomplete implementation |
| Use of Tools | 20 | Effective and advanced use of tools/technologies with proper configuration | Appropriate use of tools with correct execution | Limited or inconsistent use of tools | Incorrect or minimal use of tools |
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation | Basic analysis with limited insights | Poor or incorrect analysis of results |
| Documentation | 10 | Clear, well-structured documentation with diagrams, code, and results | Organized documentation with necessary details | Basic documentation with missing details | Poor or incomplete documentation |

# 1. Concept Map: Security Attacks, Threats, Risks, and Security Goals

## a) Relationship between Security Attacks, Threats, Risks, and Security Goals

Security attacks are attempts to compromise a computer system or network. These attacks create security threats, which lead to risks affecting an organization's information and services. Security goals—Confidentiality, Integrity, and Availability (CIA)—are implemented to reduce or eliminate these risks.

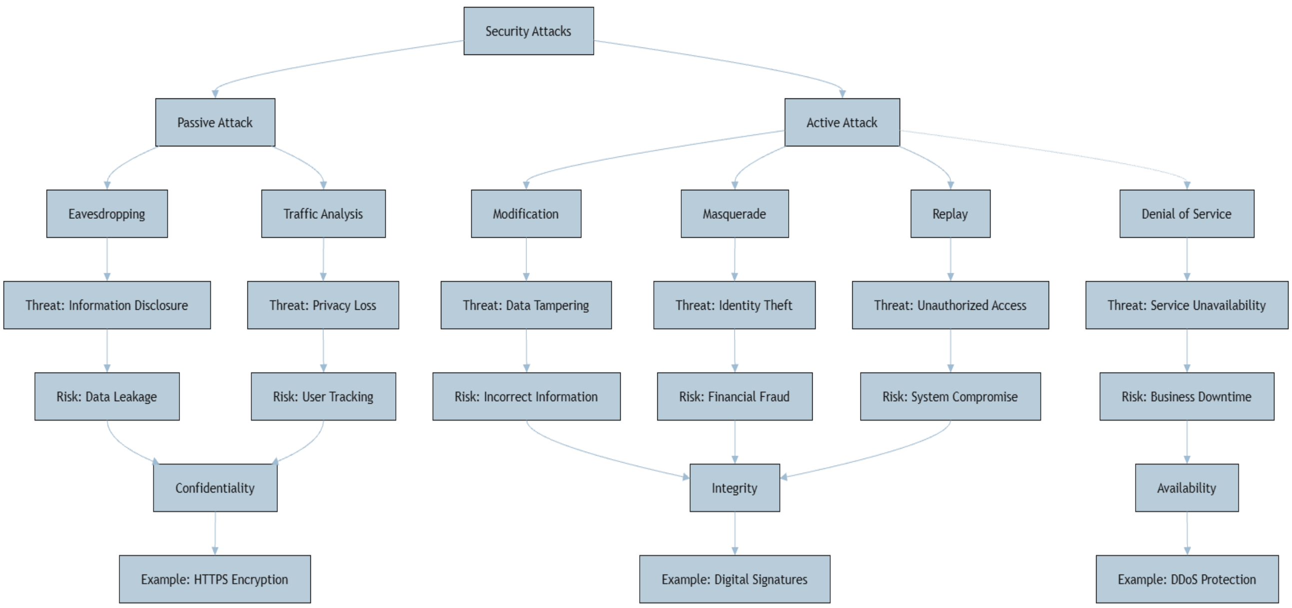
### Concept Relationships

- **Interception Attack** ○ **Threat:** Unauthorized access to information ○ **Risk:** Confidential information is exposed ○ **Security Goal: Confidentiality** ○ **Mitigation:** Encryption, authentication, access control ○ **Real-world Example:** A hacker captures passwords using packet sniffing on public Wi-Fi.
- **Modification Attack** ○ **Threat:** Unauthorized alteration of data ○ **Risk:** Incorrect or corrupted information ○ **Security Goal: Integrity** ○ **Mitigation:** Hash functions, digital signatures, checksums ○ **Real-world Example:** An attacker changes the amount in an online banking transaction. ○
- **Interruption Attack** ○ **Threat:** Loss of system or service availability ○ **Risk:** Business operations are interrupted ○ **Security Goal: Availability** ○ **Mitigation:** Backups, redundancy, DDoS protection ○ **Real-world Example:** A DDoS attack makes an e-commerce website unavailable. ○
- **Fabrication Attack** ○ **Threat:** Creation of fake data or identities ○ **Risk:** Unauthorized access and fraud ○ **Security Goal: Confidentiality and Integrity** ○ **Mitigation:** Multi-factor authentication, digital certificates ○ **Real-world Example:** A phishing email pretending to be from a bank.

### Conclusion

The CIA triad protects systems from different types of attacks by ensuring that information remains confidential, accurate, and available to authorized users.

## Concept Map




## 2. Concept Map: Comparison of Substitution Techniques

Substitution ciphers encrypt plaintext by replacing characters with other characters according to a specific rule or key.

| Cipher | Key Usage | Complexity | Security Strength |
|---|---|---|---|
| Caesar Cipher | Single shift key | Low | Weak |
| Monoalphabetic Cipher | Fixed substitution alphabet | Medium | Moderate |
| Vigenère Cipher | Keyword-based multiple alphabets | Medium | Strong |
| Hill Cipher | Matrix key | High | Very Strong |

### Analysis

#### Caesar Cipher

- Uses a single shift value.
- Easy to implement and break using brute force.

#### Monoalphabetic Cipher

- Uses one fixed substitution alphabet.
- More secure than Caesar but vulnerable to frequency analysis.

#### Vigenère Cipher

- Uses a keyword to apply multiple substitution alphabets.
- Reduces the effectiveness of frequency analysis.

#### Hill Cipher

- Uses matrix multiplication for block encryption.

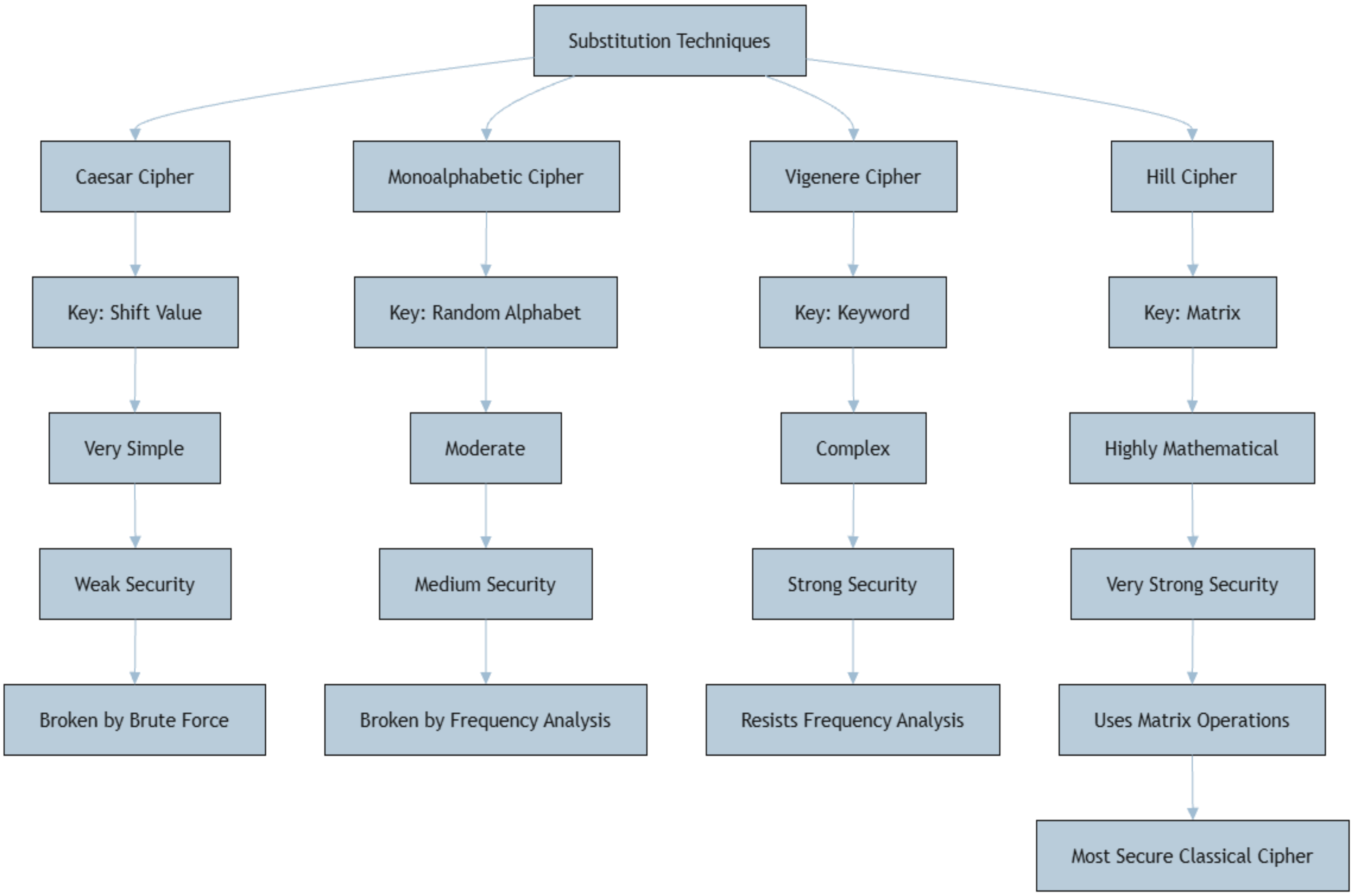
- Encrypts multiple letters together.
- Stronger against frequency analysis than the other classical substitution ciphers.

### Most Secure Technique

**Hill Cipher** is the most secure among these techniques because:

- It encrypts blocks of characters instead of one character at a time.
- Matrix operations increase complexity.
- Frequency analysis becomes much more difficult.

# Concept Map



## 3. Concept Map: Classical Encryption Schemes

Classical encryption techniques are divided into **Substitution Techniques** and **Transposition Techniques**.

### Similarities

- Both convert plaintext into ciphertext.
- Both use a secret key.
- Both aim to protect confidential information.

### Differences

| Feature | Substitution Technique | Transposition Technique |
|---|---|---|
| Working Principle | Replaces characters | Rearranges character positions |
| Data Structure | Character values change | Character positions change |
| Character Frequency | Changes symbols | Frequency remains the same |
| Security Level | Moderate | Moderate |

### Effect on Data Structure

- **Substitution** ○ Characters are replaced with different symbols.
    - ○ Original positions remain unchanged.
- **Transposition** ○ Characters remain the same.
  ○ Their positions are rearranged.

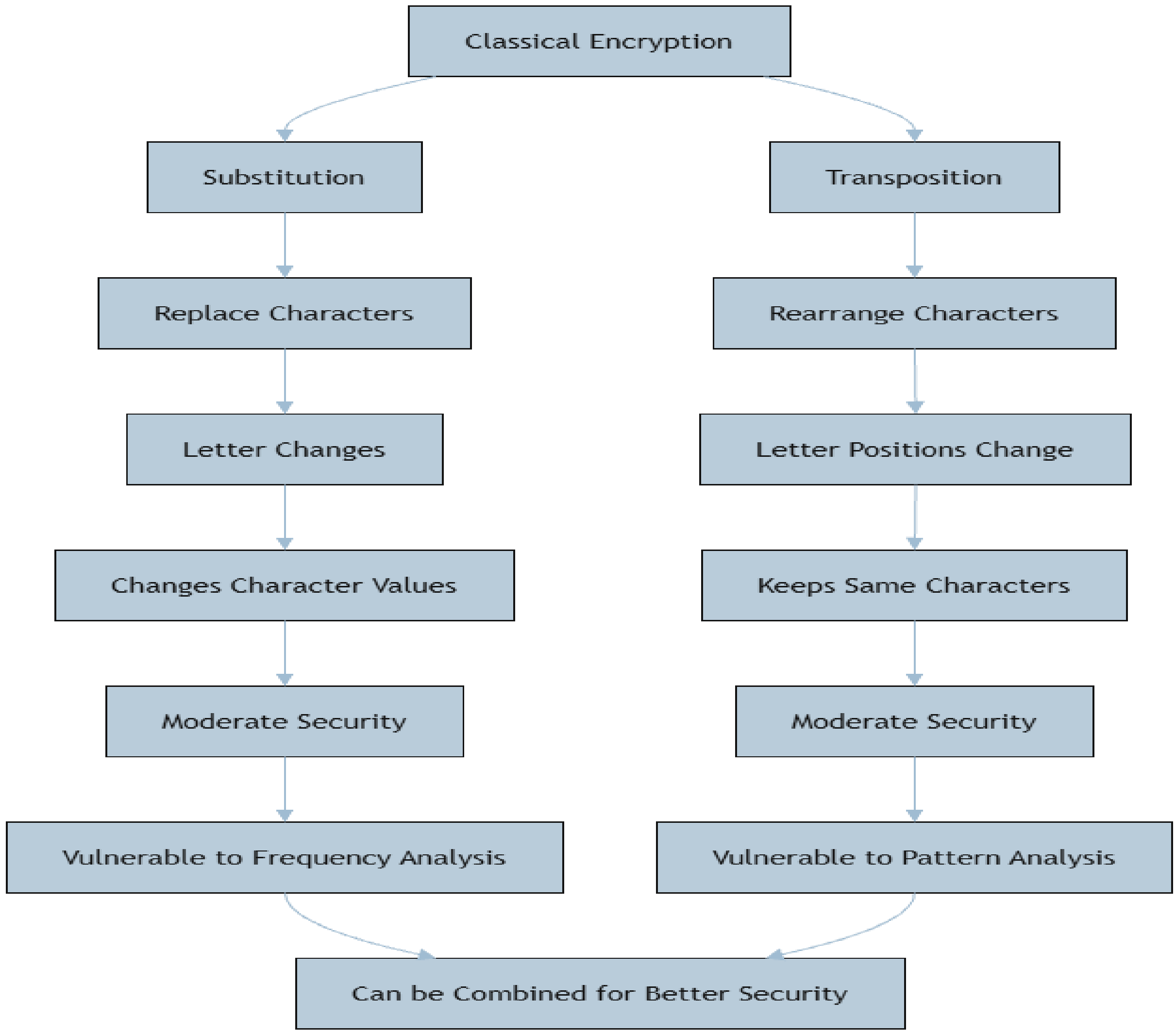
### Security Analysis

- **Substitution Ciphers** ○ Vulnerable to frequency analysis because letter occurrence patterns remain.
- **Transposition Ciphers** ○ Vulnerable to pattern analysis and brute-force attacks. ○ Usually more resistant to frequency analysis than substitution.

#### More Vulnerable

**Substitution techniques** are generally more vulnerable because attackers can use letter frequency statistics to identify plaintext characters.

## Concept Map




## 4. Concept Map: Steganography

Steganography is the practice of hiding information inside digital media without revealing that communication exists.

**Relationship between Concepts**

### Data Hiding

- Conceals secret information inside images, audio, or video.
- Used to transmit confidential information secretly.

**Application:** Hiding military or intelligence messages inside digital images.

### Watermarking

- Embeds ownership information into digital media.
- Used for copyright protection and authentication.

**Application:** Copyright protection of digital photographs and videos.

### Survivability

- Ensures hidden information remains intact after compression, editing, or transmission.

- Improves reliability of hidden data.

**Application:** Medical image authentication where patient information remains protected even after image processing.

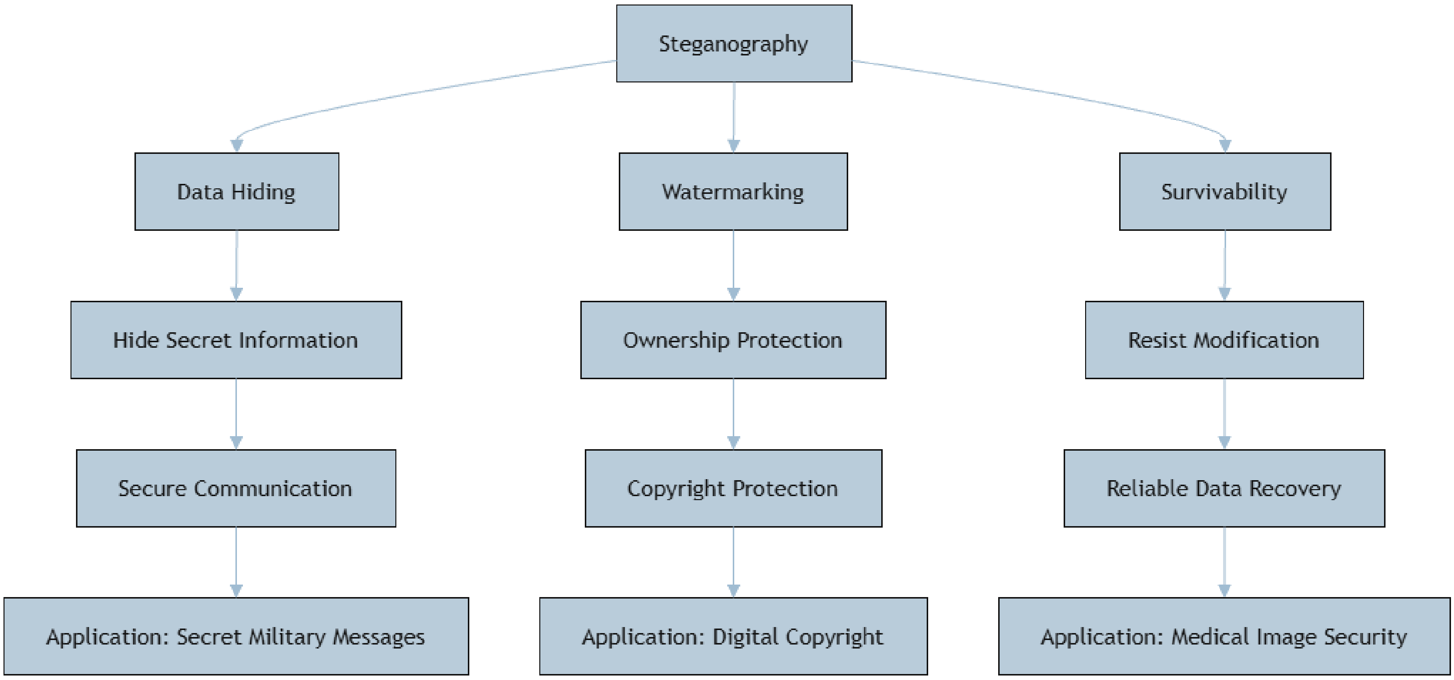
**Contribution to Secure Communication**

| Concept | Contribution |
|---|---|
| Data Hiding | Keeps secret communication invisible |
| Watermarking | Verifies ownership and authenticity |
| Survivability | Protects hidden data from damage or modification |

### Conclusion

These three concepts together improve secure communication by hiding information, proving ownership, and ensuring the hidden information survives various modifications.

# Concept Map




## 5. Concept Map: Number Theory in Cryptography

Number theory forms the mathematical foundation of modern public-key cryptography.

**Mathematical Connections**

### Extended Euclid Algorithm

- Computes the Greatest Common Divisor (GCD).

- Finds the Modular Inverse.

### Modular Inverse

- Calculates the multiplicative inverse under modular arithmetic.
- Used to generate RSA private keys.

### Fermat's Little Theorem

- Simplifies modular exponentiation.
- Used in prime number-based cryptographic calculations.

### Euler's Phi Function (φ(n))

- Counts numbers relatively prime to n.
- Required for RSA key generation.

### Euler's Theorem

- Extends Fermat's Little Theorem.
- Provides the mathematical basis for RSA encryption and decryption.

## Role in Encryption and Decryption

| Concept | Role |
|---|---|
| Extended Euclid Algorithm | Finds modular inverse |
| Modular Inverse | Generates private key |
| Fermat's Little Theorem | Supports modular arithmetic |
| Euler's Phi Function | Calculates RSA keys |
| Euler's Theorem | Enables RSA encryption and decryption |

## Most Critical Concept

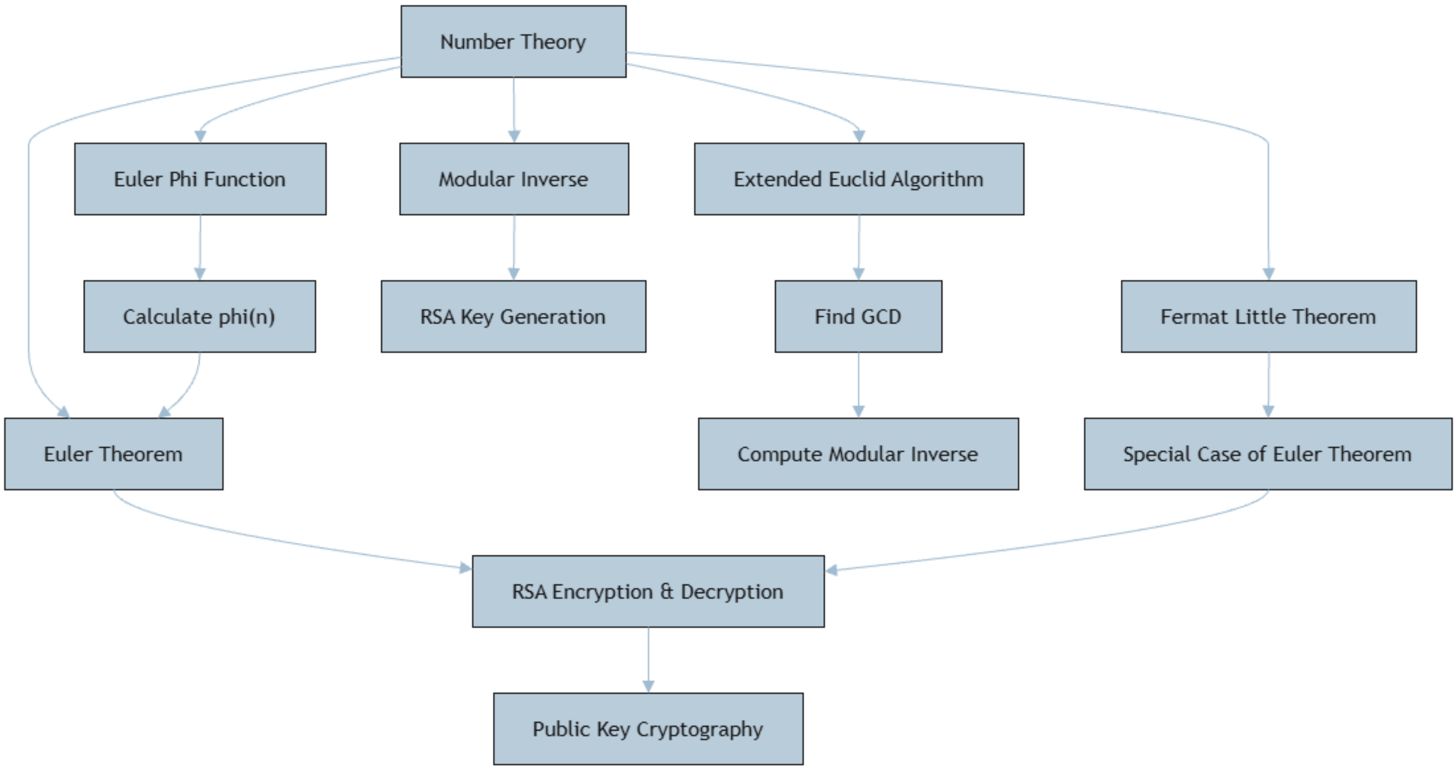
**Euler's Phi Function (φ(n))** is the most critical concept for public-key cryptography because:

- It is essential for RSA key generation.
- It determines the relationship between public and private keys.
- It is used to compute the modular inverse.
- Without φ(n), RSA encryption and decryption cannot be performed correctly.

### Conclusion

Number theory concepts work together to provide secure encryption, key generation, and decryption in public-key cryptographic algorithms such as RSA.

# Concept Map


Number Theory
Euler Phi Function
Modular Inverse
Extended Euclid Algorithm
Calculate phi(n)
RSA Key Generation
Find GCD
Fermat Little Theorem
Euler Theorem
Compute Modular Inverse
Special Case of Euler Theorem
RSA Encryption & Decryption
Public Key Cryptography